7.6 Hyperbolic Rotations and Relativity

WORKSHEET

The question

STEP III 1991, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s3-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What identity do $\cosh u$ and $\sinh u$ satisfy, analogous to $\cos^2 u + \sin^2 u = 1$? And what are the addition formulas for $\cosh(u+v)$ and $\sinh(u+v)$?

STAGE 1 The hyperbola is invariant

Covers: the hyperbola, and the lines through the origin mapped to themselves

Ask yourself Expand $x'^2 - y'^2$. The cross terms cancel. Which identity finishes it off?

Pause and try A line through the origin is mapped to itself when its direction is an eigenvector. Try the directions $\begin{pmatrix}1\\1\end{pmatrix}$ and $\begin{pmatrix}1\\-1\end{pmatrix}$.

STAGE 2 Adding hyperbolic angles

Covers: the gradients, $v' = v + u$

Ask yourself A line with gradient $\tanh v$ has direction $(\cosh v, \sinh v)$. Apply T to this direction and use the addition formulas. Why does it not matter whether the line passes through the origin?

STAGE 3 When are angles preserved?

Covers: the condition for equal angles

Ask yourself Use $\tan\psi = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$ with $m_i = \tanh v_i$, and multiply top and bottom by $\cosh v_1 \cosh v_2$.

Pause and try You should find $\sinh(v_1 - v_2)$ on top and $\cosh(v_1 + v_2)$ underneath. Which of these changes under T? When is $\cosh(v_1 + v_2) = \cosh(v_1 + v_2 + 2u)$?

STAGE 4 **The picture**

Covers: the lines that meet at the origin

Ask yourself When $v_1 + v_2 = -u$, what are v_1' and v_2'? Compare the new lines with the reflections of the old ones in the x-axis.

What's really going on

Special relativity

In Einstein's special relativity (1905), different observers moving at constant velocity relative to each other disagree about time and distance, but they all agree on the quantity $t^2 - x^2$ (with the speed of light $c = 1$), the *spacetime interval*. The transformations preserving it are the *Lorentz transformations*, and in one space dimension they are exactly the matrices in this question, with x as time and y as distance. An observer moving with velocity $v = \tanh u$ uses coordinates related to ours by this matrix, with $\cosh u = \frac{1}{\sqrt{1-v^2}}$, the famous factor γ of time dilation. Hermann Minkowski saw in 1908 that this makes spacetime a four-dimensional geometry with a "distance" that has a minus sign in it: "Henceforth space by itself, and time by itself, are doomed to fade away into mere shadows."

What the question means physically

- **The invariant lines** $y = \pm x$ are the paths of light rays. They are the same for every observer: the speed of light is the same for everyone, the founding principle of relativity.
- **The hyperbola** $x^2 - y^2 = 1$ is the set of events one unit of "proper time" from the origin, as measured by observers moving at all possible constant velocities.
- **Adding rapidities.** A line of gradient $\tanh v$ through the origin is the path of an object moving at velocity $\tanh v$. A boost by rapidity u changes its velocity to $\tanh(v + u) = \frac{\tanh v + \tanh u}{1 + \tanh v \tanh u}$, which is Einstein's velocity-addition formula from Section 3.8. Velocities do not add; rapidities do.

And in Section 5.8 we saw the same addition formula as the distance rule in the Poincaré

disc: the set of possible velocities in relativity, measured with rapidity, is a hyperbolic space.

Try this next

1. **Composing boosts.** Show that $T(u)T(w) = T(u+w)$, where $T(u)$ is the matrix in the question. What is $T(u)^{-1}$?

2. **Time dilation.** A spaceship travels at 0.8 times the speed of light. Find its rapidity u and $\cosh u$. If 10 years pass on Earth, how much time passes on the ship? (Use $t_{\text{ship}} = t_{\text{Earth}} / \cosh u$.)

3. **Hyperbolic area.** Show that the area enclosed by the hyperbola $x^2 - y^2 = 1$, the x-axis and the line from the origin to $(\cosh v, \sinh v)$ is $\frac{v}{2}$, just as the corresponding area for a circle is $\frac{\theta}{2}$. Why must T preserve such areas?

4. **Squeeze.** Change coordinates to $p = x + y$, $q = x - y$. Show that T becomes $p \mapsto e^{u}p$, $q \mapsto e^{-u}q$, and that the hyperbola becomes $pq = 1$. Describe the transformation geometrically in these coordinates.

Further reading

- **Edwin Taylor & John Wheeler, *Spacetime Physics*** (W. H. Freeman). Special relativity with rapidity and spacetime diagrams, for students with A-level mathematics.
- **Tevian Dray, *The Geometry of Special Relativity*** (CRC Press). Relativity as hyperbolic trigonometry, in exactly the spirit of this section.
- **Albert Einstein, *Relativity: The Special and the General Theory*** (various editions). Einstein's own popular account, still very readable.

The full guided solution is in Section 7.6:
https://beyondstep.keithmcnulty.org/ch07/hyperbolic-rotations.html